Detailed Project Approach, Timeline, and Implementation Strategy

1. Recommended Implementation Strategy

Business-First Agile Approach

Our strategy centers on delivering immediate business value while building a scalable foundation for Duke Energy Residential Solutions' long-term growth. We propose a hybrid agile methodology that combines rapid iteration with enterprise-grade architecture planning.

Core Strategic Pillars:

- Customer-Centric Development: Start with existing Duke customers' most critical pain points
- Parallel Workstream Execution: Simultaneous development of mobile app, APIs, and backend systems
- Incremental Value Delivery: Bi-weekly demos with usable features every sprint
- Risk-Mitigated Rollout: Phased deployment starting with pilot contractor network

2. Critical Assumptions Affecting Project Approach

Technical Assumptions

- Duke Energy will provide API documentation and sandbox access within 2 weeks of project kickoff
- Existing Commerce/Dynamics systems can support API load without performance degradation
- Contractor portal systems have available integration endpoints or can be modified
- SpeedPay integration specifications are available and current

Business Assumptions

- Duke SMEs will be available 10-15 hours per week for requirements validation and testing
- Contractor network will participate in pilot program (current assumption: 25 contractors across 3 markets)
- Product catalog for ad-hoc services will be defined by Duke during Weeks 8-12
- Customer communications (legal/privacy notices) will be approved by Duke legal team

Resource Assumptions

- Duke IT will provide integration support and security reviews within agreed timeframes
- Customer research access will be granted for user testing phases
- Content for DIY library will be provided or jointly developed during design phase

3. Functional and Technical Risks

HIGH RISK - Requires Immediate Attention

- Legacy System Integration Complexity: Commerce/Dynamics systems may have undocumented limitations
  - *Mitigation:* Early integration prototyping, dedicated integration specialist, fallback API gateway
- Contractor Network Adoption: Slow contractor onboarding could limit service availability
  - *Mitigation:* Dedicated contractor success manager, phased rollout, incentive program
- Payment Processing Compliance: Multiple payment flows (utility billing, credit cards, contractor collection)
  - *Mitigation:* Early legal/compliance review, PCI-DSS certification planning, payment flow prototyping

MEDIUM RISK - Monitor Closely

- Data Privacy Regulations: Home inventory data sensitivity across multiple states
    - *Mitigation:* Privacy-by-design architecture, legal review checkpoints, data minimization strategies
- Mobile App Store Approval: Complex payment flows may trigger additional app store scrutiny
    - *Mitigation:* Early app store consultation, alternative payment flow designs, web app fallback
- Scalability Planning: Rapid user growth could overwhelm initial architecture
    - *Mitigation:* Cloud-native design, auto-scaling infrastructure, performance testing throughout development

LOW RISK - Standard Mitigation

- Third-party API Dependencies: SpeedPay or other vendor API changes
- Resource Availability: Key Duke Energy team member availability during critical phases
- Technology Evolution: Mobile OS updates during development cycle

4. Work Outside Project Scope

Explicitly Excluded

- Utility Billing System Integration: Direct integration with regulated utility billing systems
- Contractor Business System Modifications: Changes to individual contractor management software
- Marketing Campaign Development: Customer acquisition campaigns and promotional materials
- Legal Documentation: Terms of service, privacy policies, regulatory compliance documentation
- Content Creation: Photography, video production for DIY library (consultation provided)

Future Phase Considerations

- IoT Device Integration: Smart home device connectivity (Phase 2/3)
- AI/ML Features: Predictive maintenance algorithms (Phase 2)
- White-label Platform: Multi-tenant architecture activation (Phase 3)
- Advanced Analytics: Business intelligence dashboards (Phase 1.5)

5. Transition from RFP to Project Launch

Contract Negotiation Phase (2-3 weeks)

- Week 1: Technical deep-dive sessions with Duke IT and business teams
- Week 2: Integration testing with Duke systems, security review initiation
- Week 3: Final scope refinement, resource allocation confirmation, contract execution

Project Initiation (Week 1 post-contract)

- Day 1-2: Team onboarding, security clearances, system access requests
- Day 3-5: Discovery workshop completion, requirements finalization
- Week 1 End: Project charter approval, communication protocols established

Early Success Milestones

- Week 2: First API connection established
- Week 4: Interactive prototype demonstration
- Week 8: MVP feature set validation with pilot users

6. Detailed Project Plan by Phase

PLANNING PHASE (Weeks 1-4)

Key Activities:

- Stakeholder interviews and requirements validation
- Technical architecture design and approval

- User research and persona validation
- Project infrastructure setup

Deliverables:

- Finalized requirements document
- Technical architecture specification
- User experience strategy
- Project communication plan

ANALYSIS & HIGH-LEVEL DESIGN (Weeks 4-12)

Key Activities:

- API specification development
- Database schema design
- User interface wireframing
- Integration mapping

Deliverables:

- API documentation and specifications
- Database design document
- UX wireframes and user journeys
- Integration architecture document

DETAILED DESIGN (Weeks 8-16)

Key Activities:

- High-fidelity UI mockups
- Detailed technical specifications
- Security and compliance framework
- Testing strategy development

Deliverables:

- Complete UI/UX design system
- Technical specification documents
- Security implementation plan
- Comprehensive test plan

BUILD PHASE (Weeks 12-32)

Key Activities:

- Agile development sprints (2-week iterations)
- Continuous integration/continuous deployment setup
- Regular stakeholder demonstrations
- Ongoing code review and quality assurance

Deliverables:

- MVP release (Week 20)
- Functional Alpha (Week 24)
- Beta version (Week 28)
- Release candidate (Week 32)

TEST PHASE (Weeks 24-36)

Key Activities:

- Comprehensive testing execution
- User acceptance testing coordination
- Performance and security testing
- Bug resolution and optimization

Deliverables:

- Test execution reports
- Performance benchmarking results
- Security audit completion
- Go-live readiness assessment

DEPLOYMENT (Weeks 36-40)

Key Activities:

- Production environment setup
- Phased rollout execution
- Go-live support
- Initial user onboarding

Deliverables:

- Live production system
- User training materials
- Operational procedures
- Performance monitoring setup

WARRANTY PERIOD (Weeks 40-44)

Key Activities:

- Post-launch monitoring and support
- Issue resolution and optimization
- Performance tuning
- Knowledge transfer

Deliverables:

- Stability reports
- Performance optimization recommendations
- Complete documentation handover
- Transition to ongoing support

7. Project Timeline Summary

Phase	Duration	Start	Finish	Key Milestone
Planning & Analysis	12 weeks	Week 1	Week 12	Requirements Finalization
Design & Architecture	8 weeks	Week 8	Week 16	Design Approval

Development MVP	8 weeks	Week 12	Week 20	MVP Release
Alpha Development	4 weeks	Week 20	Week 24	Alpha Testing
Beta Development	4 weeks	Week 24	Week 28	Beta Release
Final Development	4 weeks	Week 28	Week 32	Release Candidate
Testing & QA	4 weeks	Week 32	Week 36	Go-Live Approval
Deployment	4 weeks	Week 36	Week 40	Production Launch
Warranty Support	4 weeks	Week 40	Week 44	Transition Complete

Total Project Duration: 44 weeks (11 months)
Target Go-Live: Q2 2026 (aligns with Duke's Q2-Q3 2026 target) **
(** dependant on project start date)

8. Resource Requirements and Skill Sets

Orases Team Structure (Vendor-Provided)

Core Team -

- Technical Product Manager: 10+ years enterprise integration experience
- Project Manager: PMP certified, enterprise software experience
- (2) Senior Full-Stack Developers: Enterprise app experience
- (1) Full-Stack Developer: React/Node.js, API development, database design
- Senior Business Analyst: Requirements gathering, process optimization

Specialized Support (Part-time/Phase-specific) -

- Product Designer: Mobile-first design, enterprise user experience
- DevOps Engineer: Cloud infrastructure, CI/CD, security implementation
- QA Specialist: Test automation, performance testing

Peak Resource Period: Weeks 16-32 (Development phase)

Duke Energy Required Resources

Business Team -

- Product Owner: Requirements validation, business decision authority
- Business Analyst: Process documentation, contractor coordination
- Marketing Representative: Brand guidelines, content approval
- Process SMEs: As needed

Technical Team (if available) -

- IT Architect: Integration planning, security review
- Database Administrator: Data migration support, performance optimization
- Security Specialist: Compliance validation, security testing